Speaker Notes
AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media

Diploma defence speech notes

Bachelor's diploma defence, 2026

Contents

Slide 1. AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media

Purpose:

Open the defence, introduce the team, and set the research-prototype scope.

What to say:

Good morning. The topic of our diploma work is *AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media*. The work was prepared by Nurshanov Dias, Askarov Insar, Kassymkhanov Timur, and Kaidrakhmetova Dinara under the supervision of Dr. Bakhyt N. Alipova. The main idea is to study how artificial intelligence can support directional prediction of thermoelastic wave behaviour in simplified geological media. From the start, I would like to define the scope clearly: this is a research prototype and a comparative framework. It is not a field-calibrated geophysical simulator and it is not proposed as a replacement for full numerical modelling.

Key points:

- The defence concerns AI-assisted prediction of thermoelastic wave propagation.
- The task is source–probe directional prediction in simplified geological media.
- The result is a comparative research framework, not an industrial simulator.

Transition to next slide:

I will begin with the motivation and aim of the work.

Slide 2. Motivation and Aim

Purpose:

Explain why the topic matters and state the aim without overclaiming the prototype.

What to say:

Temperature changes in rocks can affect stress, deformation, and wave propagation. This is relevant for geothermal systems, underground engineering, and geophysical interpretation, where thermal and mechanical effects can interact. The aim of the thesis is to develop and justify an AI-guided framework for directional prediction of thermoelastic wave propagation in geological media under simplified effective-medium assumptions. The important word here is "framework". We are not trying to model a real field site with all geological complexity. Instead, we create a controlled environment where different AI surrogate models can be compared under the same source–probe prediction task.

Key points:

- Thermal changes can create mechanical effects in rock.
- The aim is a directional prediction framework under simplified assumptions.
- The scope is a research prototype, not a field-validated simulator.

Transition to next slide:

Next, I will show the geological media and effective parameters used in the model.

Slide 3. Geological Media We Model

Purpose:

Show how geological materials enter the prototype through effective scalar parameters.

What to say:

This slide shows representative effective parameters for four geological media: granite, basalt, sandstone, and limestone. The table includes density, compressional wave velocity, thermal conductivity, heat capacity, and thermal expansion where available. These values are used as effective scalar parameters under a locally homogeneous isotropic approximation. This means the model does not explicitly resolve pores, cracks, bedding, fluids, or anisotropy. The diagram on the right shows the simplified two-dimensional plane-strain domain with a thermal source point and a probe point. The line between them defines the direction of prediction.

Key points:

- Rock types differ in mechanical and thermal properties.
- The prototype uses effective isotropic material parameters.
- The source–probe geometry defines the directional query.

Transition to next slide:

After defining the materials, I will introduce the coupled thermoelastic system.

Slide 4. Coupled Thermoelastic System

Purpose:

Present the theoretical backbone of the work in a compact, defensible way.

What to say:

The coupled thermoelastic system combines three ideas. First, the constitutive law relates stress to elastic strain and temperature perturbation. The term $\gamma = 3K\alpha$ represents thermoelastic coupling, where K is the bulk modulus and α is thermal expansion. Second, the equation of motion links stress gradients to acceleration and displacement. Third, the heat equation describes the evolution of temperature and includes coupling with volumetric deformation. In physical terms, heat changes temperature, temperature produces thermal strain, strain contributes to stress, and stress drives displacement and wave-like response.

Key points:

- Stress contains elastic and thermal contributions.
- Motion is driven by stress gradients.
- Heat transfer and deformation are coupled through the thermoelastic term.

Transition to next slide:

Now I will translate this physical model into the two-dimensional directional prediction task.

Slide 5. 2D Directional Prediction Task

Purpose:

Define the practical prediction query and make the mathematical scope explicit.

What to say:

The practical task is two-dimensional directional prediction. The model domain is $\Omega \subset \mathbb{R}^2$, and we assume plane strain, small deformation, and linear isotropic thermoelasticity. The primary fields are temperature $T(x,t)$ and in-plane displacement $u(x,t) = (u,v)$. A source point x_s and a probe point x_p define the vector $d = x_p - x_s$, the distance L, the unit direction $\hat{d}$, and the azimuth angle ϕ. The model mapping $\mathcal{F} : X \to Y$ means that material, geometry, and time inputs are mapped to a predicted response. The scope is directional response, not complete field-scale geological simulation.

Key points:

- The task is defined by a source point and a probe point.
- Distance, unit direction, and azimuth are explicit geometric quantities.
- The prediction maps material, geometry, and time inputs to selected outputs.

Transition to next slide:

To train and compare the models, the work uses a controlled synthetic dataset.

Slide 6. COMSOL-Based Synthetic Dataset

Purpose:

Explain where the reference data come from and why they are synthetic.

What to say:

The synthetic dataset was generated in COMSOL Multiphysics using a coupled thermoelastic model. The setup is a two-dimensional plane-strain geological domain with a size of $1\,\mathrm{m} \times 1\,\mathrm{m}$. The initial temperature is $293.15\,\mathrm{K}$ and the source temperature is $1500\,\mathrm{K}$. From this model we export temperature, displacement, stress, strain, and velocity fields. These are not field measurements. They are controlled numerical reference data, which makes them suitable for comparing several surrogate models under the same assumptions.

Key points:

- The dataset is COMSOL-derived and synthetic.
- The domain is a simplified 2D plane-strain setting.
- Exported fields provide reference data for surrogate comparison.

Transition to next slide:

The next slide shows the mesh and numerical domain used for these exports.

Slide 7. COMSOL 2D Model and Mesh

Purpose:

Show the numerical domain and explain why mesh structure matters.

What to say:

This slide shows the two-dimensional COMSOL model and finite-element mesh. The domain represents a simplified section of geological medium with an internal heated source. The mesh is refined near the source because this is where the strongest temperature, displacement, and stress gradients are expected. This mesh is important because it defines the spatial nodes and local structure from which the COMSOL fields are exported. Later, this same structure is useful for mesh-aware modelling, especially for the MeshGraphNet route.

Key points:

- The simulation is a simplified two-dimensional domain.
- Mesh refinement is concentrated around the thermal source.
- The mesh provides the spatial structure for exported fields.

Transition to next slide:

Now I will show the main simulated thermoelastic fields.

Slide 8. COMSOL Simulation Results

Purpose:

Connect the synthetic thermal source with temperature, displacement, and stress outputs.

What to say:

The three images show the main result fields from the COMSOL setup. The first image is the temperature field: local heating creates a high-temperature region near the source while the surrounding rock remains closer to the initial temperature. The second image shows total displacement caused by thermal expansion. The third image shows von Mises stress, with stress concentration near the heated region. Together, these fields demonstrate the coupled thermal and mechanical response used as the reference basis for the neural surrogate models.

Key points:

- Local heating produces a localized temperature disturbance.
- Thermal expansion creates displacement.
- Stress concentration confirms coupled thermoelastic behaviour in the synthetic setup.

Transition to next slide:

After simulation, the exported fields must be represented in a model-ready format.

Slide 9. Data Representation and Preprocessing

Purpose:

Explain how COMSOL exports are converted into inputs for different model families.

What to say:

The COMSOL simulation results are exported as CSV files for each field and time step. These exports are aligned into a common node-based representation. Dynamic fields are assembled into a spatio-temporal tensor $\mathbf{S} \in \mathbb{R}^{N_t \times N_p \times F_d}$, where N_t is the number of time steps, N_p is the number of spatial points, and F_d is the number of dynamic field channels. Static material and geometry features are stored separately. Before training, each feature channel is normalized with the usual mean and standard deviation formula. This gives a common processed dataset for coordinate-based, grid-based, graph-based, and attention-based routes.

Key points:

- CSV exports are aligned by nodes and time steps.
- Dynamic fields form a spatio-temporal tensor.
- Normalization makes model training and comparison more stable.

Transition to next slide:
Next, I will describe the comparative prototype framework.

Slide 10. Comparative Prototype Framework

Purpose:
Explain that all model routes are evaluated under one shared workflow.

What to say:
The prototype uses one workflow for all model routes. The frontend, backend orchestrator, material catalog, and model services follow one shared input–output contract. This is important for fair comparison: each route receives the same source–probe geometry, the same material presets, and the same normalized response format. The goal is not to present a production platform. The goal is to make a consistent research environment where different AI-guided prediction methods can be compared under the same conditions.

Key points:

- All routes use comparable inputs.
- Outputs are normalized into the same response structure.
- The framework supports fair model comparison.

Transition to next slide:
Before discussing individual models, I will define the shared comparison contract.

Slide 11. Shared Model Comparison Contract

Purpose:
Clarify the inputs, outputs, and interpretation shared by all surrogate routes.

What to say:
All routes are treated as surrogate modelling approaches for the same AI-guided thermoelastic prediction task. The input side includes material parameters, source–probe geometry, observation time, and processed field features. The output side includes estimated temperature, displacement components, travel time, and diagnostic response quantities. The comparison basis is the same material presets, geometry contract, response format, and analytical travel-time sanity check. This keeps the comparison fair and avoids presenting the models as complete numerical solvers.

Key points:

- The comparison uses one shared prediction contract.
- Inputs and outputs are kept consistent across models.

- The routes are surrogate models, not equivalent physical solvers.

Transition to next slide:
I will start the model discussion with the physics-informed baseline.

Slide 12. PINN: Physics-Informed Baseline

Purpose:
Introduce the PINN route and explain why it is the main physical baseline.

What to say:
PINN stands for Physics-Informed Neural Network. In this framework, it is the explicitly physics-informed route because its training objective can include thermoelastic residuals in addition to supervised data terms. It predicts temperature and displacement fields and serves as the physics-informed baseline for comparison with more data-driven routes. The interpretation must still be balanced: PINN is a neural surrogate whose loss encourages physical consistency. It is not an exact numerical solver and it is still limited by the simplified two-dimensional source–probe setup.

Key points:

- PINN combines supervised data with physical residual terms.
- It predicts temperature and displacement fields.
- It is the strongest physics-informed baseline in the comparison.

Transition to next slide:
The next slide shows the PINN input contract and architecture.

Slide 13. PINN Input and Architecture

Purpose:
Explain what the PINN receives, what it predicts, and why the architecture is differentiable.

What to say:
The public neural contract maps coordinates, time, and material parameters to temperature and displacement outputs. The input includes coordinates, time, Young's modulus, Poisson's ratio, density, thermal expansion, conductivity, and heat capacity. The output includes temperature and displacement components. The architecture can be viewed as an MLP-based route, with a more developed version using separate coordinate and material encoders, a fused residual trunk, and separate heads for temperature and displacement. Smooth, differentiable outputs are important because physics residuals require spatial and temporal derivatives through automatic differentiation.

Key points:

- Inputs are geometry, time, and material properties.
- Outputs are temperature and displacement components.
- Differentiability allows residuals to be evaluated from derivatives.

Transition to next slide:
With the architecture defined, I will explain the composite loss.

Slide 14. PINN Composite Loss and Residuals

Purpose:

Show how data fitting and physical consistency are combined in the PINN route.

What to say:

The PINN loss combines supervised field loss, velocity consistency, elastic wave residuals, and coupled thermal residuals. The supervised term compares predicted fields against COMSOL-derived reference data. The velocity term supports dynamic interpretation from time derivatives. The wave and thermal residuals encourage consistency with the thermoelastic equations introduced earlier. This is the main reason PINN is treated differently from purely supervised baselines. At the same time, the slide also states a limitation: dedicated boundary-condition and initial-condition losses are not yet enforced, so the result must still be validated carefully.

Key points:

- The loss combines supervised, velocity, elastic, and thermal terms.
- Residuals are computed using automatic differentiation.
- Boundary and initial losses are a current limitation.

Transition to next slide:

The second model family is the Fourier Neural Operator.

Slide 15. Fourier Neural Operator

Purpose:

Introduce FNO as the structured-grid neural operator route.

What to say:

The Fourier Neural Operator is included as a field-to-field neural operator. Instead of predicting only a scalar output, it learns a mapping from an input field space to an output field space. In the practical two-dimensional setup, the input tensor has batch, channel, and two spatial dimensions, and the output tensor has the same spatial grid with selected output channels. For the three-channel thermoelastic setup, the outputs are temperature and two in-plane displacement components. FNO is useful because it can exploit regular grid structure, but in the current prototype it remains a supervised neural-operator branch rather than an explicitly physics-informed solver.

Key points:

- FNO learns a mapping between spatial fields.
- The practical setup uses two-dimensional tensors.
- The output channels represent temperature and in-plane displacement.

Transition to next slide:

The next slide shows the main FNO processing pipeline.

Slide 16. FNO Pipeline

Purpose:

Summarize the processing stages of the FNO route.

What to say:

The FNO pipeline maps an input field to an output field through several stages. First, a lifting layer maps input channels into latent features. Then spectral blocks process the latent field. Finally, a projection layer maps the latent representation back to physical output channels. Inside the spectral block, the model uses Fourier transform, retained modes with learned weights, inverse Fourier transform, and a local convolution branch. In simple terms, the spectral part helps capture broader field patterns, while the local branch preserves nearby spatial corrections.

Key points:

- FNO uses lifting, spectral processing, and projection.
- Fourier operations capture global spatial structure.
- Local convolution helps represent short-range details.

Transition to next slide:

Now I will explain the spectral block more directly.

Slide 17. FNO Spectral Block

Purpose:

Explain the Fourier-space operation and retained modes at an intuitive level.

What to say:

The spectral block updates the latent field by combining a spectral branch and a local convolution branch. The spectral branch transforms the field into Fourier space, applies a learned spectral weight tensor to selected coefficients, and returns the result to physical space using the inverse transform. Only a limited number of low-frequency modes are retained. This reduces computational cost and controls model capacity. The important interpretation is that FNO learns global and local field structure from data, but these operations do not directly enforce the thermoelastic PDE residuals.

Key points:

- The spectral branch works in Fourier space.
- Retained modes control spectral resolution and cost.
- The block is data-driven, not a PDE residual calculation.

Transition to next slide:

The next slide gives the training objective and the current caveat for FNO.

Slide 18. Training and Evaluation of FNO

Purpose:

State how FNO is trained and make the current calibration limitation transparent.

What to say:

The FNO route is trained with a supervised objective combining mean squared error and relative error. Normalization is computed on the training set and reused during validation, testing, and inference. Evaluation includes visual field comparison, channel-wise errors, relative L^2 error, and physical sanity checks. The most important caveat is that the current FNO prototype still requires output-scale recalibration before it can be treated as a physically reliable predictor. Therefore, in the results section it should be interpreted as a diagnostic branch rather than as a final calibrated model.

Key points:

- FNO training uses supervised MSE and relative error terms.
- Channel normalization is reused consistently.
- The current output scale needs recalibration.

Transition to next slide:

The third model family is MeshGraphNet.

Slide 19. MeshGraphNet: Graph Formulation

Purpose:

Introduce MeshGraphNet as the mesh-aware surrogate route.

What to say:

MeshGraphNet represents the COMSOL computational mesh as a graph. The mesh nodes form the set $\mathcal{V}$ and neighbouring connections form the edge set $\mathcal{E}$. Each node contains coordinates, current physical state, material parameters, and scenario conditions. Each edge contains relative geometry, such as the vector and distance between neighbouring nodes. This is useful because COMSOL data are naturally mesh-based. MeshGraphNet can preserve nonuniform mesh connectivity more naturally than a regular grid model, although it is still a supervised surrogate rather than a full physical solver.

Key points:

- The mesh is represented as nodes and edges.
- Node features carry state, material, and scenario information.
- Edge features carry local geometric relations.

Transition to next slide:

The next slide shows the input and output structure in more detail.

Slide 20. MeshGraphNet: Inputs and Outputs

Purpose:

Clarify what is stored at nodes and edges and what the graph model predicts.

What to say:

This slide separates node features, edge attributes, and output fields. Nodes store coordinates, material parameters, scenario condition, and current dynamic state. Edges store relative spatial relations between neighbouring nodes. The output is the predicted next physical field vector for each mesh node. This makes MeshGraphNet suitable for state-transition style prediction. The key distinction from PINN is that physical quantities enter through features and training data, while the governing equations are not enforced as explicit residuals in the same way.

Key points:

- Nodes contain geometry, material data, and physical state.
- Edges describe local spatial relationships.
- The model predicts next physical fields over mesh nodes.

Transition to next slide:

Now I will summarize the graph neural network architecture.

Slide 21. MeshGraphNet: Encoder–Processor–Decoder

Purpose:

Describe the MeshGraphNet architecture without excessive implementation detail.

What to say:

The architecture has three main parts. The encoder maps raw node and edge features into latent vectors. The processor applies repeated message-passing blocks over the graph connectivity. The decoder maps the final latent node states into predicted physical fields. In this implementation the model uses a latent dimension of 128, ten message-passing steps, MLP encoders, and a node-wise decoder. This design is common in neural simulators because it separates feature embedding, learned interaction modelling, and output prediction.

Key points:

- Encoder: embeds raw node and edge features.
- Processor: performs repeated message passing.
- Decoder: converts latent node states to physical outputs.

Transition to next slide:

The next slide explains the message-passing block itself.

Slide 22. MeshGraphNet: Message Passing Block

Purpose:

Connect graph message passing with local physical interaction learning.

What to say:

In the message-passing block, edge representations are updated using information from neighbouring node states and current edge attributes. Then incoming edge messages are accumulated for each destination node using scatter-add aggregation. Finally, the node state is updated with a residual connection and layer normalization. This can be interpreted as a learned local interaction mechanism on the mesh. It is useful for modelling spatial dependencies, but physical reliability still depends on validation against reference fields because the equations are learned implicitly rather than enforced as residuals.

Key points:

- Edge updates exchange information between neighbouring nodes.
- Aggregation collects incoming messages at each node.
- Physical behaviour is learned from data and must be validated.

Transition to next slide:
The fourth model family is the Transformer-style surrogate.

Slide 23. Transformer: Inputs and Outputs

Purpose:
Introduce the attention-based route and its token representation.

What to say:
The Transformer route treats sampled points as tokens. Input tokens include coordinates, initial temperature and displacement state, velocity-like components, and material parameters. Query tokens contain the coordinates where fields are requested. The output at each query point is temperature and two in-plane displacement components. This token structure makes the route flexible because the number of input points and query points can differ. Its advantage is fast query-style prediction, while its limitation is weaker explicit physics enforcement compared with PINN.

Key points:

- Input tokens contain state, geometry, and material information.
- Query tokens specify prediction locations.
- Outputs are temperature and in-plane displacement at query points.

Transition to next slide:
Next, I will show how the Transformer architecture processes those tokens.

Slide 24. Transformer: Architecture

Purpose:
Summarize the encoder–decoder attention mechanism for query prediction.

What to say:
The Transformer architecture has an encoder and a decoder. The encoder processes input tokens using self-attention layers, building a representation of the known thermoelastic state. The decoder processes query tokens and uses cross-attention to attend to the encoded input representation. Finally, a linear head returns the predicted temperature and displacement components. In intuitive terms, each requested query point can use attention to decide which input points are most relevant for its prediction. This supports flexible query-point inference, but attention is still a learned statistical mechanism rather than an explicit thermoelastic conservation law.

Key points:

- Encoder self-attention processes known input tokens.

- Decoder cross-attention connects query points with encoded inputs.
- The output head predicts (T, u, v) at requested points.

Transition to next slide:

The next slide defines the training loss for this route.

Slide 25. Transformer: Training Loss

Purpose:

Explain the supervised objective and its limited physical regularisation.

What to say:

The Transformer loss combines a supervised reconstruction term, a velocity consistency term, and a thermal residual term. The supervised loss reconstructs the COMSOL reference fields. The velocity consistency term aligns displacement rate with exported velocity. The thermal residual encourages diffusion-like behaviour through thermal diffusivity $k/(\rho C_p)$. The training uses AdamW and token subsampling with $N_{\text{in}} = N_q = 1024$ to control computational cost. The interpretation remains cautious: this is mainly a supervised attention-based surrogate, not a strict physics-informed solver like the PINN route.

Key points:

- The objective combines supervised, velocity, and thermal terms.
- Token subsampling controls attention cost.
- Physics constraints are weaker than in PINN.

Transition to next slide:

Now I will move from model descriptions to calculation status and results.

Slide 26. Calculation Status

Purpose:

Summarize the evaluation setup and state caveats before showing plots.

What to say:

The calculation setup uses forty scenarios per material and four materials: granite, sandstone, limestone, and basalt. With four model routes, this gives $40 \times 4 \times 4 = 640$ model responses under one shared contract. The slide also emphasizes zero fallback: every plotted response is an active model output, not a mock result. The operating envelope remains two-dimensional plane strain. The FNO caveat is repeated here because it affects interpretation: FNO is included in the comparison, but its output scales are still miscalibrated, so it should currently be treated as a diagnostic branch.

Key points:

- The comparison covers 640 model responses.
- Plotted outputs come from active models.
- FNO is included but currently miscalibrated.

Transition to next slide:

The first result slide compares predicted travel times.

Slide 27. Travel-Time Predictions

Purpose:

Explain what the travel-time plot shows and how to interpret the model differences.

What to say:

This plot compares predicted travel time by model and material. The approximate values shown on the slide are about 0.11 ms for PINN, 0.14 ms for Transformer, 1.29 ms for FNO, and 2.47 ms for MeshGraphNet. The key observation is that model-family differences are larger than the basalt–sandstone material difference in the current prototype. This does not mean one model is universally correct. It means that model formulation strongly affects directional travel-time estimates, so comparison and validation are necessary before drawing physical conclusions.

Key points:

- The plot compares travel-time estimates across models and materials.
- Model-family differences dominate the material difference here.
- The result motivates careful validation.

Transition to next slide:

To evaluate accuracy, the next slide compares predictions with an analytical reference.

Slide 28. Accuracy Against Analytical Reference

Purpose:

Present the main quantitative result and its cautious interpretation.

What to say:

Here the predicted travel times are compared with a simple analytical reference, $t_{\mathrm{gt}} = r/V_p$. In this formula, t_{gt} is reference travel time, r is source–probe distance, and V_p is compressional wave velocity. This is not a full thermoelastic validation; it is a sanity check under a simple isotropic approximation. In the balanced sweep, PINN has a median error of about 2.0% and remains within the 10% band in all cases. Transformer has a larger median error, while FNO and MeshGraphNet show much larger deviations in the current setup.

Key points:

- The analytical reference is a sanity check, not complete validation.
- PINN shows the best agreement with the reference.
- Other routes need calibration or stronger time conditioning.

Transition to next slide:

Accuracy is one side of the comparison; inference latency is also important.

Slide 29. Inference Latency and Time Behavior

Purpose:

Compare practical response time and explain the fixed-horizon issue.

What to say:

The latency plot shows that all routes return predictions in milliseconds on the development machine, which is fast enough for interactive prototype use. The median latency is approximately 1.3 ms for Transformer, 1.5 ms for MeshGraphNet, 2.4 ms for FNO, and 2.6 ms for PINN. However, speed is not the only criterion. PINN is also the only route that changes smoothly with requested observation time. The other routes currently behave more like fixed-horizon predictors, so their temporal behaviour needs additional development for time-dependent directional studies.

Key points:

- All routes are fast enough for interactive prototype comparison.
- Transformer has the lowest median latency.
- PINN currently has the most meaningful time-conditioned behaviour.

Transition to next slide:

Before concluding, I will state the main threats to validity.

Slide 30. Threats to Validity

Purpose:

Show scientific caution and define the limits of the results.

What to say:

The main threats to validity are listed here. First, the training data are limited and come from a pilot COMSOL-derived configuration, not from a full calibrated geological database. Second, the physical setup is simplified to two-dimensional plane strain with effective material parameters. Third, the current FNO route has a normalization and output-scale issue. Fourth, the analytical reference r/V_p is useful but isotropic and incomplete for full thermoelastic validation. These limitations are important because the results should be interpreted only within the simplified assumptions of the study.

Key points:

- The dataset is limited and synthetic.
- The physical formulation is simplified.
- The analytical reference is useful but limited.

Transition to next slide:

After the limitations, I will briefly show the web application prototype.

Slide 31. Web Application Prototype

Purpose:

Show that the thesis includes a usable local MVP for comparing model routes.

What to say:

This slide shows the implemented web application prototype. The local workspace connects a native web frontend, a FastAPI backend orchestrator, the geological material catalog, and separate model routes for MeshGraphNet, FNO, Transformer, and PINN. The user can choose a geological medium, configure the planar source and probe points, set the observation time, choose a model route, and inspect the normalized prediction response. The screenshots show three parts of the interface: the landing workspace, the source–probe visualization, and a detailed prediction result. The important interpretation remains the same: this is a thesis-demo interface for reproducible comparison, not a field-validated thermoelastic simulator.

Key points:

- The application connects frontend, backend, material catalog, and model services.
- It supports source–probe setup and model-route comparison.
- Results are displayed as normalized thermal, mechanical, directional, and temporal responses.

Transition to next slide:

With the prototype shown, I can summarize the conclusions.

Slide 32. Conclusions

Purpose:

State the main scientific outcomes of the diploma work.

What to say:

The thesis develops a physically grounded comparative framework for AI-guided prediction of thermoelastic wave propagation in geological media. Within the tested assumptions, PINN is the strongest physics-informed baseline and shows the best agreement with the analytical travel-time reference. Transformer is the fastest route and is useful for rapid query-style inference. MeshGraphNet remains a useful mesh-aware surrogate candidate. FNO should currently be treated as a diagnostic branch until output normalization is corrected. Overall, the work demonstrates a careful prototype for comparing neural surrogate approaches rather than a final geophysical simulator.

Key points:

- The main contribution is a comparative AI-assisted framework.
- PINN performs best against the travel-time sanity check.
- Other routes remain useful candidates with specific limitations.

Transition to next slide:

Finally, I will outline the most important next steps.

Slide 33. Future Work

Purpose:

Show how the prototype can be developed after the diploma.

What to say:

The first future direction is to retrain and recalibrate the two-dimensional FNO route. The second is to add time-conditioned rollout behaviour to MeshGraphNet. The third is to extend the material catalog with more complete thermoelastic parameters and validation data. The fourth is to compare against additional analytical or numerical baselines. The final message is that the prototype estimates directional characteristics of thermoelastic wave propagation within simplified assumptions and provides a basis for further research.

Key points:

- Recalibrate FNO.
- Improve time-conditioned behaviour in MeshGraphNet.
- Extend material data and validation baselines.

Transition to next slide:

This completes the main presentation.

Slide 34. Thank You

Purpose:

Close the presentation and invite questions.

What to say:

Thank you for your attention. I will be happy to answer your questions about the physical assumptions, the COMSOL dataset, the model comparison, or the interpretation of the results.

Key points:

- Close politely.
- Signal readiness to discuss assumptions and limitations.
- Keep answers consistent with the research-prototype framing.

Transition to next slide:

Questions from the committee.

Slide 35. Backup: Mechanical Rock Parameters

Purpose:

Use this backup only if the committee asks about mechanical material values.

What to say:

This backup table gives the extended mechanical material catalogue. It includes density, Young's modulus, Poisson's ratio, shear modulus, bulk modulus, and representative P-wave and S-wave velocities for ten rock types. I would not read every value. The main point is that the prototype uses physically meaningful effective parameters, while still treating them as midpoint values under isotropic assumptions.

Key points:

- Mechanical parameters control inertia, stiffness, and wave speeds.

- Values are effective and comparative, not site-calibrated.
- The table supports the material assumptions used in the main flow.

Transition to next slide:
If needed, the next backup slide gives thermal and porosity parameters.

Slide 36. Backup: Thermal and Porosity Parameters

Purpose:
Use this backup if asked how thermal properties and porosity enter the catalogue.

What to say:
This table gives thermal expansion, thermal conductivity, heat capacity, volumetric heat capacity, thermoelastic coupling, and porosity information. These quantities control how a temperature disturbance spreads and how temperature change contributes to thermal strain and stress. Dashes indicate values that were not available in the source catalogue. They are left as missing values rather than filled by unsupported assumptions.

Key points:

- Thermal properties control heat spreading and thermoelastic strain.
- Porosity is included as an effective descriptive property.
- Missing values are shown explicitly.

Transition to next slide:
If the question concerns wave velocities, use the next backup slide.

Slide 37. Backup: Elastic Wave Velocities

Purpose:
Explain the P-wave and S-wave formulas if the committee asks about wave physics.

What to say:
This slide gives the standard formulas for compressional and shear wave velocities. P-waves involve particle motion mainly parallel to the propagation direction and include volume change. S-waves involve transverse motion and depend on shear rigidity. The formulas show that stiffness increases wave velocity, while density increases inertia. These relations also explain why the analytical reference $t = r/V_p$ is a useful first sanity check for travel time.

Key points:

- V_P depends on bulk modulus, shear modulus, and density.
- V_S depends on shear modulus and density.
- These formulas justify the simple travel-time reference.

Transition to next slide:
If the question concerns heat-spreading time scale, use the next backup slide.

Slide 38. Backup: Thermal Diffusivity

Purpose:

Explain the compact thermal parameter used to interpret heat spreading.

What to say:

Thermal diffusivity is defined as $\alpha_T = k/(\rho C_p)$. It has units of square metres per second and describes how quickly a temperature disturbance spreads through a material. Larger thermal diffusivity means faster temperature smoothing. This quantity is useful because it combines conductivity, density, and heat capacity into one interpretable parameter for the thermal part of the coupled response.

Key points:

- Thermal diffusivity describes heat-spreading rate.
- It combines conductivity, density, and heat capacity.
- It helps interpret the time scale of the thermal response.

Transition to next slide:

The final backup slides document image credits and references.

Slide 39. Image Credits

Purpose:

Document visual sources if the committee asks about images.

What to say:

This slide lists the sources for the external educational diagrams and rock textures used in the presentation. It helps separate illustrative visual material from the thesis results. The COMSOL field images and model-result plots are part of the project workflow, while the external diagrams are used only to support physical explanation.

Key points:

- External images are credited separately.
- Illustrative diagrams are not model outputs.
- Project-generated figures are distinct from source images.

Transition to next slide:

The reference slide gives the scientific and technical sources behind the work.

Slide 40. References

Purpose:

Point to the scientific basis of the thesis if asked.

What to say:

The references include rock physics, thermoelasticity, geodynamics, surface geophysics, anisotropy, neural operators, and Transformer-related sources. In the defence, I would use this slide only in the question period. It supports the theoretical background, material-property assumptions, and machine-learning model choices used throughout the thesis.

Key points:

- References cover physics, geology, and neural surrogate modelling.
- They support the theory and material assumptions.
- The slide is mainly for Q&A support.

Transition to next slide:

End of backup material.

Possible questions from the committee

1. Why did you choose directional prediction instead of full-field simulation?

Because the thesis focuses on a narrower and more defensible research task. Full-field thermoelastic simulation is more computationally and physically complex. Directional prediction allows us to compare source–probe responses, travel time, and model behaviour under controlled assumptions.

2. Is the prototype validated on real field data?

No. The current work uses COMSOL-derived synthetic data and analytical sanity checks. The results should be interpreted within simplified modelling assumptions. Field data would be an important next step, but it is outside the current scope.

3. Why is PINN called a physics-informed baseline?

PINN is called a physics-informed baseline because its training objective includes residual terms connected with thermoelastic relations. This means that physical structure is included in the learning objective, but the result is still interpreted as a baseline model inside the simplified prototype.

4. Why does FNO perform poorly in the current results?

The current two-dimensional FNO route has a calibration and output-scale issue. Therefore, it is kept in the comparison as a diagnostic branch. Its architecture is still relevant, but this implementation needs retraining and normalization correction.

5. What is the role of the analytical reference $t = r/V_p$?

It is a simple travel-time sanity check based on distance and P-wave velocity. It is not a complete thermoelastic validation, but it helps identify whether predicted travel times are physically plausible in a basic isotropic approximation.

6. Why is Transformer fast but less physically reliable?

The Transformer route uses attention-based query prediction and can return estimates quickly, but most of its behaviour is learned from data. It has weaker explicit physical constraints than PINN, so its predictions require careful validation.

7. What is the current limitation of MeshGraphNet?

MeshGraphNet preserves mesh topology and is suitable for mesh-aware surrogate modelling, but in the current prototype it behaves mainly as a fixed-horizon predictor. It needs stronger time-conditioned rollout behaviour for more reliable time-dependent prediction.

8. What are the main limitations of the work?

The main limitations are the simplified two-dimensional setting, effective isotropic material parameters, limited synthetic data, fixed-horizon behaviour in some surrogate routes, the FNO normalization issue, and the absence of direct calibration against real geological measurements.

9. What is the main contribution of the diploma work?

The main contribution is a physically grounded comparative framework for AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media. It combines thermoelastic theory, geological material parameters, a controlled dataset, and several neural surrogate routes under one consistent comparison workflow.

Short version of the speech

Good morning. The topic of our diploma work is *AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media*. The work presents a research prototype for AI-guided directional prediction of thermoelastic wave propagation in simplified geological media. It is not a field-calibrated simulator. The goal is to estimate selected directional characteristics and compare several neural surrogate approaches under one shared contract.

The physical motivation is that temperature changes in rocks can create thermal expansion, stress, displacement, and wave-like mechanical response. This is relevant for geothermal systems, underground engineering, and geophysical interpretation. A full coupled numerical simulation can be expensive when many materials and source–probe directions must be compared, so the thesis formulates a narrower directional prediction task.

The model uses effective geological material parameters such as density, elastic moduli, thermal conductivity, heat capacity, thermal expansion, and P-wave and S-wave velocities. The formulation is two-dimensional, assumes plane strain, small deformation, and linear isotropic thermoelasticity, and does not explicitly resolve pores, fluids, fractures, plasticity, or full anisotropy.

Directional prediction is defined by a source point and a probe point. Their difference gives the direction, distance, and azimuth. The prototype estimates temperature, displacement, travel time, and directional response for that query. The data are based on a controlled COMSOL-derived two-dimensional thermoelastic setup, exported and normalized for several model families.

The comparison includes a Physics-Informed Neural Network, Fourier Neural Operator, MeshGraphNet, and Transformer-style surrogate. PINN is used as the physics-informed baseline because its objective includes thermoelastic residual terms. FNO, MeshGraphNet, and Transformer-style models are treated as surrogate routes for comparison, not as physically equivalent numerical solvers.

The calculations use forty scenarios per material, four materials, and four model routes, giving six hundred forty model responses. In the travel-time comparison, model-family differences are larger than the material difference in the current prototype. Against the simple analytical reference $t = r/V_p$, PINN shows the best agreement and remains within the ten percent band in the balanced sweep. Transformer-style prediction is the fastest route, while MeshGraphNet and FNO need additional development for time behaviour and calibration.

The main limitations are the limited synthetic dataset, the simplified two-dimensional assumptions, the current FNO normalization issue, fixed-horizon behaviour in some routes, and the fact that the analytical reference is only a sanity check. The conclusion is that the work provides a physically grounded comparative framework for AI-guided thermoelastic wave prediction within simplified assumptions. Future work should recalibrate FNO, improve time-conditioned MeshGraphNet behaviour, extend material data, and compare against additional analytical or numerical baselines.